Statistics

★ Modules

1. Basic Probability & Stat
2. Conditional Prob & Baye's thm
3. Prob Distributions.
4. Hypothesis Testing
5. Prediction & Forecasting
6. Gaussian mixture model & Expectation maximization.

★ Books

T1 Statistics for Data Scientists → Maurits Kaptein → Springer 2022

T2 Probability & Statistics for Engineering & Sciences → Jay Devore → Cengage 8th edition

T3 Intro to Time series & forecasting → 2nd edition → Brockwell & Davis springer
↳ M5

★ Evaluation Components

EC 1	→ Quiz 1 & Quiz 2	Online	10%.
	→ Assignment 1 & 2	Online	20%.
EC 2	→ Mid sem exam	Closed Book	30%.
EC 3	→ Comprehensive exam	open Book	40%.

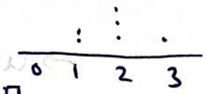
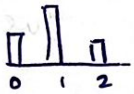
★ Contact Session 1

→ Measures of Central Tendency & Measures of Variability
→ Data - Symm & Asymm
→ outlier detection
→ 5 point summary

★ Stats used to

→ collect data
→ systematically org data
→ andlyse data
→ infer about data
→ Take decision from data.

★ Representation Formats

→ Tally Tables 𝍸1 6
→ Dot Diagram

→ Bar Chart

→ Freq Distributions

	Freq	CD
10-20	3	3
20-30	5	8

→ Histograms

→ Cumulative Distribution (CD)

excel
↓
=randbetween(1,75)
gives random no. b/w 1 & 75

Σx_i
= summation of all values till now
= always → starts 0
↳ ends 1

★ Data Types

Data
- Categorical (Qualitative)
  - Nominal
    - discrete unordered
    - → unmarried / married
    - → country of residence
    - → colours
    - To summarise use freq or % proportions
  - Ordinal
    - discrete ordered
    - → Feedback scale (strongly disagree, disagree, neutral ....)
    - → use freq to summarise proportions sometimes means
- (Interval/Ratio) Numerical (Quantitative)
  - Discrete Σ (counted)
    - → no of pets
    - → no of packages
    - use summation to summalise
  - ∫dx Continuous (measured)
    - → airplane speed
    - → time spent
    - use integration to summarize
    - Interval: no zero pt
    - Ratio: zero pt

(2)

★ Levels of Data Measurement

① Nominal scale

-> classifies data into distinct categories
-> no ranking implied.

eg. marital status, country of residence, colour
->freq, mode

② Ordinal scale

-> distinct categories
-> ranking implied

eg. satisfaction, grades, medals
-> mode, median, frequencies.

-> unequal interval
(we cant assume dist b/w agree & strongly agree is same as neutral & agree)

③ Interval scale

-> ordered scale

equal interval-> diff b/w measurement is meaningful quantity
-> measurement doesnt have true zero point

eg. temp in °F & °C
months of year (no month called zero)
time of day

④ Ratio scale

-> ordered scale

equal interval -> diff b/w measurements is meaningful quantity
-> measurements have true zero point

eg. weight, Age, Salary, time duration

★ True zero point = can you say something is twice or half as much

↑ more mathematical power

Ratio
→ ordered, equal intervals, true zero
↳ can compare differences & ratios

Interval
→ ordered, equal intervals, ✗ true zero
↳ can compare differences only

Ordinal
→ ordered, unequal intervals
↳ freq., mode, median

Nominal
→ unordered = just names or labels
↳ frequencies, mode

↓ less mathematical meaning

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★ Measures of Central Tendency:

→ convenient way of using single number that describes performance of the group.

→ single value is used to describe center of data

→ 3 common measures

⓪ sum $= \Sigma x_i$

① Mean $= \Sigma x_i / n$ = balancing pt → true measure

i/2 ② Median = mid pt after sorting → (based on count)
↳ not affected by outlier

③ Mode = most repeating values
↳ physical interpretation =
if there are duplicates, mean shifts towards repeating values. (dominant)

★ Mean (Arithmetic Average)

→ $\bar{Y} = \frac{\Sigma Y_i}{N}$

→ For grouped data $\bar{Y} = \frac{\Sigma f_i Y_i}{N} = \frac{\text{score} \times \text{freq}}{\text{total no.}}$

Properties

→ measures stability = every score contributes to mean

→ easily affected by outliers

→ sum of each score's distance from mean = 0

$$\Sigma (Y_i - \bar{Y}) = 0$$

Σwt on right = Σwt on left

→ can be applied to interval level of measurement

When to use

→ to calculate other measures like std. deviation etc

→ sampling stability is desired

★ Mode

→ category or score with largest freq or %.
→ can be calculated for nominal, ordinal or interval-ratio

Properties

→ used for Qualitative & Quantitative data types.
→ may not be unique
→ not affected by outliers

When to use → when data measured on nominal scale

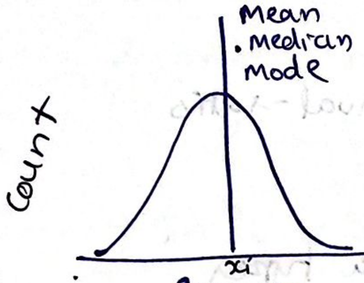
★ Median (50th percentile or middle score)

→ score that divides data into 2 equal parts
→ Total values
  ↳ odd : median = middle value of sorted data
  ↳ Even : median = avg of middle scores of sorted data

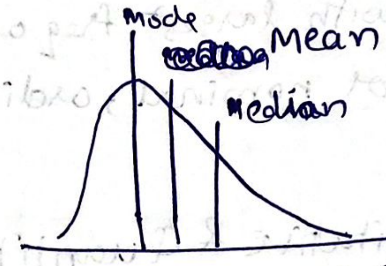
→ percentage = Mean
percentile = Median

★ Bimodal = 2 distinct modes
Multi-modal = more than 2 distinct modes.
No modal = 0 modes ⇒ distribution is straight line

★ Data: Symmetrical & Asymmetrical



Symm

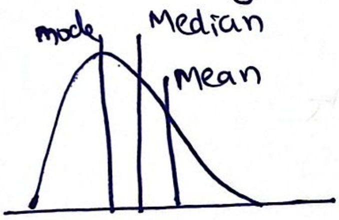


Asymm or Skewed

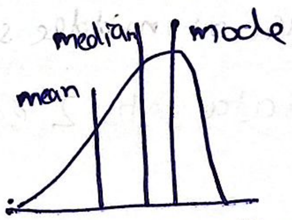
→ Peak will always be mode

→ Skew

- Negatively or Left : mean < median tail on left
- Positively or Right : mean > median tail on right



Positive skew



Negative skew

→ 50% of data left of median & 50% to the right

★ Note

→ Empirical Relationship = relationship supported by experiment & observation but not necessarily by theory

→ For moderately skewed data (skewness b/w −0.5 to 0.5) empirical relationship is

3Median = Mode + 2Mean